

The Topology of Transformation

An Exhaustive Analysis of the Nexus Framework, Configuration Cohomology, and Direct-Coupled Reader Logic

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The formal representation of dynamic systems has historically prioritized a static ontology, wherein systems are defined by their instantaneous states and governed by external algorithms that dictate chronological progression. Dean A. Kulik's theoretical architecture—specifically the Nexus framework—reverses this epistemological hierarchy. Within this paradigm, a physical or combinatorial system is defined primarily by its space of admissible transformations. The finite constructions below are not offered as toy substitutes for reality; they are explicit mathematical probes of transformation structure. Their role is to determine which consequences follow from the proposed rules without confusing the representation with the physical system it is intended to explain. Structural persistence, physical loss, and observational uncertainty are not artifacts of disappearing states, but precise geometrical manifestations of topological constraints on a continuous transformation field. This framework posits that the ultimate explanatory target of theoretical physics is not a terminal state of reality, but the mathematical rules governing transformation itself; discovering the answer key to reality's continuation does not require the termination of the processes it describes.

This analysis synthesizes Kulik's investigations into symmetric token configuration graphs, direct-coupled readers, and the algebraic topology of permutation recovery. By formalizing configuration spaces as relative cohomology rings, analyzing the information-theoretic profiles of partition readers, and mapping Eulerian transition systems to double cosets of Young subgroups via discrete Morse theory, an integrated mathematical architecture is revealed. This architecture provides mathematical realizations of several phenomena that, in physical descriptions, appear as destruction, statistical noise, or observational loss. In the Nexus interpretation, these are candidate manifestations of admissibility boundaries, monodromy in covering spaces, and invariant representation limits; the mathematical constructions below specify exactly which parts are proved within the stated model and which parts are physical interpretation.

The Algebraic Topology of Transformation and Admissibility

The foundation of the Nexus framework rests on finite token-configuration graphs. For a finite, connected substrate graph $X = (V, E)$, one may consider K -token configurations subject to exclusion rules. These finite systems are used as explicit mathematical realizations of transformation structure, not as claims that physical reality is merely a toy model. Their purpose is to determine which consequences follow from the proposed transformation rules. By allowing a distinguished vertex $m \in V$ to carry increasing capacity r , a nested sequence of configuration spaces is

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generated: $X^{(0)} \subset X^{(1)} \subset X^{(2)} \subset \dots$. The transitions between these spaces provide a controlled mathematical realization of what it means for a system's admissible structure to change.

The Five Mechanisms of Topological Modification

When a system undergoes a change of substrate, the transformation field $\mathcal{T}_X$ maps to a new field $\mathcal{T}_{X'}$. Early interpretations conflated any dimension reduction in the configuration space with the destruction of relational structure. However, rigorous chain-map analysis of substrate transformations proves that such a view is mathematically deficient. For the substrate transformations treated here, the change decomposes into five mechanisms. Within the Nexus definition of destruction, only exclusion removes a previously admissible continuation from the transformation field.

Modification Mechanism	Algebraic Operation	Topological Consequence	Structural Effect
Extension	$\mathcal{T}_X \subset \mathcal{T}_{X'}$	Relative restriction map Δ_R is surjective onto the residual. Cokernel is 0.	Adds admissible transformation pathways without generating loss.
Re-factorization	$T \mapsto T_2 T_1$	Dual pullback F_R^* is surjective.	Re-expresses a transformation path; the continuum structure survives refactorization intact.
Identification	$S_1 \sim S_2$	Merging configurations yields $\ker q_R^* = 0$.	Merges distinct representations; identifies distinctions rather than annihilating them.
Symmetry (Hiding)	$G \subset G'$	Injective pullback $R_{G'} \hookrightarrow R_G$.	Enlarged symmetry forces distinguishable transformations into identical orbits, removing visibility from the invariant sector but preserving the 1-skeleton.
Exclusion (Collision)	$S \notin \mathcal{S}_{\text{admissible}}$	Downward map yields $\ker \rho_R > 0$.	Forces a configuration out of the admissible state space entirely. Erases structural continuation.

The discovery that identification is topologically safe ($\ker q_R^* = 0$) while exclusion is strictly destructive ($\ker \rho_R > 0$) fundamentally reorients the understanding of structural loss. Loss does not occur because two variables are merged into one; loss occurs because the prior coexistence of states becomes inadmissible, effectively severing the continuation graph and forcing transformations into a null space.

The Capacity Defect and Relative Cohomology

To formalize the topology of this exclusion boundary, the transition from an older, smaller configuration space $Y_0 = X^{(r)}$ to a newly admitted space $Y = X^{(r+1)}$ introduces a relative layer of configurations, denoted $\mathcal{D}_r = V(Y) \setminus V(Y_0)$. The structural residue of this transition is captured by the relative cochain complex $\mathcal{C}^\bullet(Y, Y_0)$.

Geometrically, the entire previously admissible lower-capacity space Y_0 is collapsed into a single cone point, $*$. The relative complex is therefore equivalent to the reduced complex of the quotient space:

$$\mathcal{C}^\bullet(Y, Y_0) = \tilde{\mathcal{C}}^\bullet(Y/Y_0)$$

This cone construction ensures that the lower-capacity state space is not merely discarded but is retained as a critical boundary condition. Transformations in this relative layer are thus categorized as either internal transformations ($\mathcal{D}_r \leftrightarrow \mathcal{D}_r$) or cross transformations ($\mathcal{D}_r \leftrightarrow *$) representing pathways back into the previously admissible state space.

Symmetry further reduces this space. Let $G = \text{Stab}(m)$. For an oriented edge orbit, if a symmetry operation $g \in G$ reverses the edge such that $gS = T$ and $gT = S$, any invariant antisymmetric 1-cochain ω must satisfy $\omega(e) = \omega(ge) = \omega(-e) = -\omega(e)$. This forces $\omega(e) = 0$. Such flippable edge orbits contribute no invariant degrees of freedom and are formally deleted from the invariant quotient, becoming topological self-loops that do not alter the connectivity of the space.

The Cone-Quotient Graph Theorem

With flippable edges resolved into self-loops, the remaining transformation structure forms the symmetry quotient graph $\mathcal{Q}_r$. The dimension of the invariant residual sector, $\mathcal{D}_r$, corresponds exactly to the first Betti number of this quotient graph.

The mathematical proof relies on evaluating the cochain dimensions. Each non-flippable edge orbit supports one independent invariant antisymmetric coefficient, establishing that $\dim C^1_G(Y, Y_0) = E_Q$. Relative zero-cochains are functions on the G -orbits of new-layer vertices. Because the cone point is fixed to zero in the relative complex, $\dim C^0_G(Y, Y_0) = V_Q - 1$. A zero-cochain lies in the kernel of the coboundary operator d_0 precisely when it is constant on every connected component. Because the component containing the cone point must possess a value of zero, $\dim \ker d_0 = c_Q - 1$. By the rank-nullity theorem, $\text{rank } d_0, G^{\text{rel}} = V_Q - c_Q$. Thus, the dimension of the relative cohomology is given by:

$$\dim \mathcal{D}_r = \dim H^1_G(Y, Y_0) = \beta_1(\mathcal{Q}_r) = E_Q - V_Q + c_Q$$

The inclusion of the component term c_Q is mathematically essential, invalidating the simpler $E - V + 1$ formulation commonly applied to connected graphs. In fully saturated scenarios (where the capacity parameter reaches $K = |V| + r$), the newly admitted layer consists of a single configuration containing no vacancies. Consequently, there is no cross transformation capable of reaching the cone point. The quotient graph fragments into two isolated components ($c_Q = 2$), ensuring the Betti calculation perfectly accounts for the disconnected topology. Over 55 capacity transitions across varied substrate geometries and symmetry groups up to order 12, the empirical capacity defect precisely matched this Betti derivation, cementing the theorem and proving that terminal vanishing is governed by local symmetry rather than by a generic exhaustion of a physical transformation quantity.

The Contraction Factorization and Capacity-2 Transport

The identification of exclusion as the destructive mechanism within this framework motivated a control experiment separating the act of identifying nodes from the act of forbidding states. Under a standard graph contraction $X' \rightarrow X$, the mathematical sequence factors into two distinct topological mappings. The first map, $p: X' \rightarrow X'_0$, removes configurations that contain both endpoints of the edge being merged, representing the exclusion boundary. The second map, $q: X'_0 \rightarrow X$, performs the actual geometric identification of the two sites.

By tracking the descent of the residuals, it was proven that identification is unconditionally safe. Suppose an exact form upstairs pulls back to zero ($q^*f = d_0g'$ on X'_0). The pullback is zero on the collapsed items, which connect the two points of every fiber. Therefore, g' is constant on fibers and successfully descends to g on X , demonstrating

that merging configurations kills nothing. The true structural loss resides entirely in the collision kernel, $\ker\rho_R$. For trivial groups, this loss corresponds exactly to the Euler count of the deleted star:

$$\dim\ker\rho_R = \#(\text{removed items}) - \#(\text{removed configurations})$$

To definitively prove that admissibility boundaries are the sole cause of this loss, a capacity-2 intervention was structured. By relaxing the exclusion constraint at the merged site to permit a token capacity of 2, the collision states containing double occupancies are no longer deleted. This forms a rigorous commuting square between the upstairs configurations and the downstairs configurations.

Contraction Geometry	dimR: $X' / X_0' / X(1)$ / $X(2)$	$\ker q$ 2^*	$\ker\rho$ (Upstairs Collision)	$\ker\sigma$ (Downstairs Exclusion)	Image Status
$C_7 \rightarrow C_6$	16 / 13 / 7 / 10	0	3	3	$\ker\rho \subseteq \text{im } q_2^*$
$C_8 \rightarrow C_7$	30 / 26 / 16 / 20	0	4	4	$\ker\rho \subseteq \text{im } q_2^*$
$C_9 \rightarrow C_8$	50 / 45 / 30 / 35	0	5	5	$\ker\rho \subseteq \text{im } q_2^*$
$P_7 \rightarrow P_6$	26 / 20 / 11 / 17	0	6	6	$\ker\rho \subseteq \text{im } q_2^*$
$Q_3' \rightarrow Q_3$	93 / 83 / 60 / 70	0	10	10	$\ker\rho \subseteq \text{im } q_2^*$

In all evaluated cases, the square commutes exactly at the cochain level, and the pullback map q_2^* acts as a strict isomorphism between the exclusion kernel downstairs ($\ker\sigma$) and the collision kernel upstairs ($\ker\rho$). Within the stated capacity-2 construction, this establishes that the collision kernel is the exclusion rule's mathematical kernel, transported intact across the merge operation. Contraction itself contributes no additional kernel in the identification map; the observed defect is the relative contribution of the excluded star. In Nexus terms, destruction is therefore represented by the removal of a previously admissible continuation.

Hodge Theory, Gradients, and Circulation on Arenas

The investigation into the invariant 1-forms on these configuration arenas requires a strict delineation between two types of gradients. A "cascade gradient" requires that the cost of transformation varies across available directions, ensuring a graded response. A "Hodge gradient" implies specifically that an invariant 1-form is exact ($\omega = d\Phi$), rendering it an integrability statement. Until a theorem explicitly connects them, neither implies the other.

If a conserved scalar is attached to a current state, it cannot distinguish outgoing directions. Likewise, if symmetry forces the relevant invariant antisymmetric field to vanish on a self-reversing orbit, a scalar cannot supply the missing directional degree of freedom. By calculating the invariant Hodge decomposition, $\Omega^1 = \text{im } d \oplus \mathcal{H}^1 \oplus \text{im } \delta$, the source of transformation cost is exposed.

In extensive computational measurements across multiple arenas (including C_6 , Q_3 , and Z_3^2), the invariant harmonic space $\mathcal{H}^1_{\text{inv}}$ was confirmed to be exactly zero in all cases. The absence of an invariant harmonic contribution establishes the following mathematical consequence for the tested invariant Hodge sector: if a total transformation price is represented by an invariant Hodge 1-form, its harmonic component is zero. It may therefore be exact or coexact; the Q_3 calculation demonstrates why those cases must not be conflated. Consequently, for every closed orbit, the net path price must sum to zero. Bound, cyclic motion can possess arbitrarily high local transformation action while accumulating zero net signed debt.

However, the dichotomy between exact and harmonic forms overlooks the third component of the Hodge decomposition. On the three-dimensional cube Q_3 , the invariant sector supports a non-exact 1-form that is strictly coexact ($\text{im } \delta$). This coexact channel charges bounded orbits not by their topological winding number around a hole (as a harmonic form would), but by a pairing with the 2-cells the orbit bounds, essentially charging by enclosed area. This demonstrates that bound orbits are free unless the physical arena specifically carries an invariant coexact 1-form, in which case they must pay a measurable flux.

The underlying properties of these 1-skeleton invariants further reveal a fundamental particle-hole symmetry. For any configuration arena, the mapping $S \mapsto V(X) \setminus S$ takes a K -token configuration to an $(N - K)$ -token configuration. An allowed move $S - u + w$ translates flawlessly to $S^c - w + u$ in the complement. Because this action commutes with the complementation, the arena graphs are equivariantly isomorphic: $\mathcal{A}(X, K) \cong \mathcal{A}(X, N - K)$. Consequently, all purely 1-skeleton invariants, including the residual dimension $\epsilon(X, K)$, exhibit perfect symmetry between particles and vacancies.

Direct-Coupled Readers and the Limits of Observation

While substrate analysis dictates the topology of the transformation graph, the observer's actual interaction with a field constitutes an entirely distinct mathematical domain. A direct-coupled reader bypasses estimators, statistical noise models, and assumed variances. It applies a fixed partition function directly to a completely deterministic field, rendering the reader's output an exact logical projection of the physical state. In this environment, apparent stationarity is not an illusion suffered by an uncalibrated reader; it is an exact property of the map, occurring if and only if the system's entire orbit fails to exit a single preimage fiber.

Reversibility and Monotone Forgetting in Lattice Gases

The foundational substrate for analyzing direct-coupled logic is a one-dimensional lattice ring consisting of $N = 128$ cells. Each cell contains two 4-bit payloads representing left-moving and right-moving readers, totaling 1280 bits of informational state. Collisions between opposing readers are governed by a unimodular map over the ring Z_{16} :

$$(a, b) \rightarrow (a + b, a + 2b) \pmod{16}$$

Because the determinant of this matrix is 1 over Z_{16} , the transition is a strict bijection. Exact physical reversibility has been verified over 11,146 consecutive collisions; running the system forward and subsequently inverting the rule recovers the initial state down to the final bit. This explicitly refutes the naive assumption that colliding observers intrinsically overwrite or destroy information. Pointers may collide, but the underlying informational payload computes rather than erases.

An observer extracts information from this field via a reader $R(s, \tau)$ that averages the payload over s spatial cells and τ temporal steps, followed by exact quantization. By evaluating the log-cardinality of the distinct readings across a 4,096-step orbit, three sharply defined observational regimes manifest.

Reader Spatial Window (s)	Temporal Window (τ)	Extract ed Bits	Observational Regime	Physical Implication
1 to 8 cells	1 step	12.00	Saturated	Reader is limited solely by its sampling rate. Every possible distinction is resolved.

16 to 64 cells	4 to 64 steps	3.46 to 10.57	Partial	The underlying field's correlation structure begins to force redundant readings, costing the reader distinctions.
128 cells	256 steps (reported run)	0.00	Fibered	Apparent exact stationarity in the reported run: the reader's output has zero measured variance under the specified observable.

Furthermore, every step of resolution loss follows a strict mathematical law of monotone forgetting. Coarsening operations—whether expanding the spatial window or lengthening the temporal window, evaluated exactly or quantized—only merge existing fibers without ever cutting or scrambling them. The resolution layers form a nested filtration. This proves that forgetting a layer cannot corrupt what lies beneath it; the coarse reading is a strict functional projection of the fine reading, and the lost distinctions remain perfectly preserved in the object regardless of the observer's capacity to resolve them.

The Thermodynamics of Memory: Cost, Yield, and Keyframes

The mechanics of the direct-coupled reader clarify the energetic and informational costs of persistent observation. If priced by full volumetric re-scans, a low-resolution reader appears infinitely inefficient—touching 32,768 cell-states over time merely to extract zero bits of variance. However, applying a delta-coding framework proves that yield and cost are functionally identical. A coarse reader emitting zero bits registers zero block-changes, confirming that apparent stillness costs absolutely nothing to mathematically carry.

The critical threshold emerges when comparing delta coding to keyframe retention. A full-field state keyframe requires 1280 bits of data (256 bits of occupancy plus a 1024-bit payload seed). At the finest resolution ($s = 1, \tau = 1$), the specified delta-coding calculation reports 1428 bits of change per step. Under that coding rule and cost comparison, it is mathematically cheaper for the highly resolved reader to rescan the specified field than to retain a full state keyframe.

A crossover point occurs near $s = 32$ under the specified coding costs. Below this coarse resolution, the measured change stream is cheaper to record than a full state keyframe. Thus, in this model and encoding, spatial coarseness is the condition under which retaining a reference frame becomes coding-economical. This is an information/coding result; a thermodynamic interpretation requires the additional physical assumptions that connect logical erasure and storage costs to thermodynamic quantities.

A profound corollary arises regarding transformation keyframes. While a state keyframe encodes one static moment, a transformation keyframe (comprising the 1024-bit payload seed, the 256-bit occupancy pattern, and the 16-bit physical collision rule) regenerates the entire infinite orbit. This 1296-bit package represents a mere 1.25% overhead compared to the 1280-bit static state. Crucially, the fundamental “laws of physics” (the 16-bit collision rule) constitute only a tiny fraction of the package; 79% of the requisite information is simply the initial condition.

Because the substrate dynamics are linear over Z_{16} , the 128-step cycle of the spatial ring can be condensed into a single fixed transition matrix M . This allows states at step 10^{12} to be calculated instantaneously via matrix squaring (M^k). However, transitioning the 16-bit rule to a non-linear substitution map immediately destroys this analytic shortcut, rendering the system traversable only through step-by-step iteration. Much like the structure of cryptographic hashes such as SHA-256, where algebraic traversal is fundamentally blocked, this demonstrates that a physical process inherently owes an observer its continuous execution, but guarantees absolutely no analytic shortcut to its macroscopic future.

The Hypergeometric Information Profile of Permutations

To understand exactly what a partition reader extracts from a permutation T of N cells, the cells are divided into m equal blocks of size $s = N/m$. The reader only records the block-to-block transition matrix $C_m(T)$. The exact average information extracted from a uniformly distributed permutation follows a precise hypergeometric form. The expected mutual information is derived by calculating the entropy of the double-coset fibers generated by the Young subgroup of the partition.

The information theoretic equality proves that the bits seen ($-\log_2 \Pr(C_m = C)$) plus the bits hidden ($\log_2 |F_C|$) must sum to the total combinatorial entropy $\log_2 N!$. Averaging over the uniform distribution yields the exact mutual information:

$$I(T; C_m) = \log_2 N! - 2m \log_2 s! + m^2 E[\log_2 c!]$$

where each cell count has the hypergeometric marginal $c \sim \text{Hyp}(N, s, s)$, representing the number of marked items drawn without replacement; the expectation is taken with respect to this marginal.

For the minimally resolved 1-cut reader ($m = 2$), the transition table simplifies to a symmetric matrix dictated by a single crossing number. As $N \rightarrow \infty$, the local central limit theorem for the hypergeometric distribution reduces the extracted information to an asymptotic limit:

$$I(T; C_2) = \frac{1}{2} \log_2 \left(\frac{\pi e N}{8} \right) + o(1)$$

For $N = 128$, the total phase space is roughly 716 bits. The 1-cut observer recovers approximately 3.55 bits of information, an incredibly small fraction of the underlying complexity.

Further analysis proves severe structural blindness at low resolutions. Because the transition matrix at $m = 2$ is unconditionally symmetric, a 2-block reader is structurally incapable of distinguishing a transformation from its inverse (T from T^{-1}). Directional orientation only breaks symmetry when the resolution increases to $m \geq 3$ blocks. Furthermore, no partition reader short of absolute maximum resolution ($m = N$) can uniquely deduce the total number of cycles within the permutation. In any coarser representation, multiplication by internal block transpositions proves that the possible cycle counts form an unbroken interval of integers. The minimum of this interval corresponds directly to the number of connected components in the block-transition multigraph.

Unbalanced Transitions and Landauer's Principle

When this mathematical logic is extended to macroscopic maps that are not exact bijections (representing irreversible physical computing), the block transition table sets a strict theoretical floor on the minimum erased information. For a map $f: n \rightarrow n$ with block arrivals c_j and block sizes d_j , the minimum erasure over all compatible maps is $\frac{1}{n} \sum \varphi(c_j, d_j)$, where φ calculates the entropy of the most even possible distribution of arrivals into the available cells.

This minimum erasure is zero if and only if $c_j = d_j$ for all blocks, defining the perfectly balanced tables of permutations. For every unbalanced block, the forced excess costs at least $2/N$ bits per cell. By mapping this mathematical erasure to Landauer's principle, any asymmetric block transition guarantees a minimum physical heat dissipation proportional to this bit erasure. This creates a rigorous combinatorial lower bound on logical erasure from

the macroscopic contingency table. A thermodynamic energy bound follows only after applying the physical assumptions of Landauer’s principle to the logically irreversible operation.

Reader Agency and Cycle-Reversal Topology

When an observer attempts to recover the structural details of a permutation T using a coarse cut, they are faced with a choice of capabilities: extend the duration of observation by watching T iterate, refine the resolution of the cut, or retain a coarse resolution but exercise agency by moving the cut to view different partitions. Extensive combinatorial mapping demonstrates an absolute hierarchy between these observational capabilities, proving that agency fundamentally dominates duration.

Duration Versus Agency

For a system of $N = 8$ cells, a fixed balanced two-block reader watching all infinite iterations of T leaves 10.066 bits of information permanently hidden, and fails entirely to deduce the cycle type. Conversely, an observer restricted to a single temporal reading, but granted the agency to move the cut across all 35 balanced block combinations, leaves only 1.218 bits hidden and uniquely determines the cycle type.

Strikingly, an observer granted both movable cuts and infinite temporal duration also leaves exactly 1.218 bits hidden, dividing the S_8 space into 18,155 distinct equivalence classes. Once the observer possesses spatial agency, infinite temporal duration yields zero additional information. The passive causal quotient (Nerode equivalence) confirms this universally: watching a fixed coarse cut forever determines neither the cycle type, the number of cycles, nor the order of T .

The Z_2^k Cycle-Reversal Group and Exact Quotients

The exact mathematical nature of this 1.218-bit residual defect is governed by the cycle-reversal group. For any balanced two-block cut, the reading is completely determined by the scalar count of cells that remain in block 0. This scalar relies strictly on the undirected edge multiset of the permutation graph. Because a balanced cut only registers undirected topologies, reversing the orientation of any subset of T ’s cycles alters nothing in the observed table.

The group of such reversals is isomorphic to Z_2^k , where k is the number of cycles possessing a length of 3 or greater. The exact quotient result established in the underlying derivation applies for even $N \geq 8$: two permutations yield identical signatures across all possible balanced cuts if and only if they inhabit the exact same cycle-reversal orbit. The observer recovers the unoriented cycle structure with perfect topological fidelity but permanently loses the local orientations, isolating the 1.218 bits of missing information as precisely the unrecoverable directional arrows of the cyclic loops.

The only exceptions to this universal identity exist at $N = 4$ and $N = 6$. The algebraic swap identity $D_{uv} = a_u + a_v$ relies on integer bounding. At $N = 4$ and $N = 6$, arithmetic scaling factors (where $(N - 2)$ divides 4) permit the existence of anomalous pairings. These anomalies perfectly mimic one another across all block combinations due to block-diagonal topological symmetries where the m -cycle graph on a subset is mathematically complete (e.g., a triangle at $N = 6$). Because cycle graphs of $m \geq 4$ always omit a chord, these geometric obstructions vanish entirely for all $N \geq 8$.

Holonomy, Graph Braid Groups, and Orientation

Given that a balanced two-block reader is perfectly blind to cycle orientation, the emergence of directional awareness at $m = 3$ requires distinct mathematical treatment. When an observer utilizes 3 sectors, the transition table $\mathcal{C}(T)$ permits the construction of an antisymmetric matrix, $A = \mathcal{C} - \mathcal{C}^T$. Because T is a bijection, A possesses row and column sums of exactly zero, defining a conserved directed-circulation channel.

The dimension of this antisymmetric channel $\mathcal{A}_m$ is calculated as $(m - 1)(m - 2)/2$. This subspace is canonically isomorphic to the cycle space $Z_1(K_m; \mathbb{R})$ of the complete graph on m vertices, and it corresponds representationally to the irreducible Specht module $S^{(m-2,1,1)}$ of the symmetric group. At exactly $m = 3$, the dimension collapses to 1, and the space transforms via the strict sign representation of S_3 .

The Geometric Admissibility Constraint and Monodromy

A profound paradox emerges at $m = 3$: The sign of this scalar observable is absolutely not a function of the permutation T . Holding T and the physical boundaries of a 3-sector partition entirely fixed, permuting the abstract labels of the three sectors across all 6 arrangements yields identical absolute magnitudes but spans both positive and negative signs. No further refinement of the data can force the sign to resolve, because a mathematical function cannot yield two differing outputs for an identical input state.

The orientation sign is therefore not supplied by the transformation itself, but by the holonomy of the reader's own configuration space. Let $\mathcal{C}(X, m)$ represent the graph of all admissible partitions of the substrate X into m connected regions, navigated via single-cell transfers or geometric swaps. The labeled partitions form an S_m -covering space over the unlabeled base. A closed random walk in the base space lifts to a topological path in the covering space, generating a monodromy projection:

$$\rho: \pi_1(\mathcal{C}(X, m)) \rightarrow S_m$$

The orientation observable is the parity projection $\text{sgn} \circ \rho$.

For the explicitly constructed reader-configuration graphs, the computed monodromy is determined by the geometric admissibility rules of the substrate and not by the permutation field being observed:

- **One-Dimensional Substrates:** On the oriented ring constructions in this study, when the three sectors are constrained to remain contiguous arcs, the computed monodromy image is A_3 . The admissible moves used in the construction do not realize odd label exchanges. This is a result of the specified configuration graph; the general physical claim requires the same admissibility construction to be derived for the physical system under study.
- **Two-Dimensional and Three-Dimensional Substrates:** On the 2D torus ($Z_3 \times Z_3$) and 3D cube ($Q_3 = Z_2^3$) constructions studied here, admissible region exchanges include an odd permutation, and the computed monodromy image is S_3 . Thus both orientation signs occur in the corresponding reader phase space. The group-image statements here are results for the explicitly constructed reader configuration graphs; they are not being inferred from the general theory of graph braid groups alone.

This topological architecture intimately mirrors the application of discrete Morse theory to graph braid groups, as pioneered by Farley and Sabalka. Configuration spaces of graphs track discrete particles without collision, and the fundamental groups of these spaces define the graph braid groups. The reader's monodromy operates identically; orientation is the parity shadow of a permutation-valued topological transport. The physical transformation supplies

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the transition magnitude, but the observer’s spatial configuration governs whether a consistent lift to a specific sign is mathematically possible.

Eulerian Transition Systems and Young Subgroups

The analytical bridge connecting the macroscopic observations of partition readers to the microscopic combinatorial structures of permutations relies on the mathematics of double cosets. For a partition P mapping n cells into m blocks of sizes $d_1, \dots, d_m$, the permutations preserving the boundaries of those blocks define a Young subgroup $K = S_P \cong \prod_i S_{d_i}$. Two permutations possess identical reader transition tables if and only if they inhabit the exact same double coset $K\sigma K$.

To deduce the distribution of specific cycle types within a given double coset—the metric defining what cycle structures a coarse observer might implicitly hide—the algebraic problem must be mapped to graph theory via Eulerian transition systems. A square contingency table T generates a directed multigraph G_T , where vertices represent blocks and directed edges denote block-to-block transitions.

The Transition System Bijection and the BEST Theorem

A strict mathematical bijection exists between permutations matching table T (decorated with specific arc labelings) and the transition systems of G_T (decorated with point labelings). Under this mapping, the permutation conjugates to the successor map on the graph’s arcs. Consequently, the cycle type of the underlying permutation λ_τ corresponds identically to the multiset of closed-trail lengths in the Eulerian transition system λ_χ .

This defines the invariant equation linking double coset permutations M_T, λ to Eulerian transition systems TS_λ^T :

$$M_T, \lambda \prod_{i,j} T_{ij}! = TS_\lambda^T \prod_i d_i!$$

For the specific case of identifying single, unbroken cycles (C_n), the calculation invokes the classical BEST theorem. The classical BEST theorem gives the Eulerian-circuit count in terms of a rooted spanning-arborescence count and the vertex outdegrees. The normalization below follows the transition-system conventions used here; $t(G_T)$ denotes the directed spanning-arborescence count for the specified root:

$$M_T, n = t(G_T) \frac{\prod_i (d_i - 1)! \prod_i d_i!}{\prod_{i,j} T_{ij}!}$$

If the transition multigraph G_T is disconnected, no Eulerian transition system traversing all of its arcs as one circuit exists, so the corresponding single-cycle class count is zero.

The mathematical framework expands via polynomial-time arborescence calculations to any cycle type featuring a unique dominant length. Most notably, exact closed forms are proven for involution classes (permutations exclusively featuring 1-cycles and 2-cycles). It is proven that an involution exists within a double coset if and only if the underlying contingency table is perfectly symmetric ($T = T^T$). Consequently, a Young double coset contains an involution if and only if the coset itself is self-inverse.

Synthesis: The Architecture of Transformational Continuation

By formally integrating configuration graph cohomology, the geometric bounds of direct-coupled observation, and the Eulerian cycle tracking of double cosets, the Nexus framework solidifies its transformational paradigm. A physical state is merely a temporal cross-section of an ongoing relational transformation. In this architecture, spatial

loss and thermodynamic erasure are not arbitrary entropic decays, but precise mathematical consequences of admissibility boundaries.

When structural capacity is reduced, transformations are not annihilated; they are compressed into a relative topology (the cone point) or hidden within a mathematically verifiable invariant representation. The “price” of topological change is strictly defined by the Euler characteristic of the excluded topological star, enforcing the rule that destruction only occurs when a previously allowed continuation pathway is declared geometrically illegal.

Simultaneously, the observer does not passively record external truth, nor does it arbitrarily inject uncertainty. Observation is a geometric projection. Whether utilizing partition readers, delta-stream keyframes, or evaluating the parity holonomy of an S_m covering space, the observer merely limits the fiber of possible realities. Temporal irreversibility, spatial orientation, and unoriented cycle reversals are structural shadows enforced by the topological monodromy of the observer’s own configuration space. By elevating transformation between states above static state descriptions, the mathematics provide a candidate exact language for the constraints governing physical continuation. The Nexus objective is therefore explanatory: to recover the answer key of lawful transformation in reality, not merely to construct a sandbox description and not to require reality to reach a final static state. The finite constructions in this paper are measurements and proofs of specific structural consequences of that proposed reality-level framework; where a physical identification has not yet been derived, it is stated as interpretation rather than smuggled into the mathematics.

Selected External References

1. D. Farley and L. Sabalka, “Discrete Morse theory and graph braid groups,” *Algebraic & Geometric Topology* 5 (2005), no. 3, 1075–1109. DOI: 10.2140/agt.2005.5.1075. This establishes the general configuration-space and graph-braid framework used here; the specific reader monodromy images claimed in this paper are separate results of the constructions above.
2. T. van Aardenne-Ehrenfest and N. G. de Bruijn, “Circuits and trees in oriented linear graphs,” *Simon Stevin* 28 (1951), 203–217. This is the original source associated with the BEST Eulerian-circuit theorem used in the transition-system section.
3. R. Landauer, “Irreversibility and Heat Generation in the Computing Process,” *IBM Journal of Research and Development* 5 (1961), no. 3, 183–191. DOI: 10.1147/RD.53.0183. Landauer’s result is invoked only after the combinatorial logical-erasure bound has been established; the combinatorial calculation itself is not presented as a thermodynamic theorem.